

Approximate values and error

Every measurement is approximate — this lesson says by how much, and how to write it honestly.

LESSON 55–57

3 × 40 MIN

ALGEBRA 8, §§18–19

STAGE 9 · 3.4

LESSON CLOCK

5'

12'

8'

13'

2'

Warm-up	5 min
Explanation	12 min
Interactive	8 min
Practice	13 min
Homework	2 min

LEARNING OBJECTIVES

- Distinguish an exact value from an approximate one.
- Calculate the absolute error and the relative error.
- Round to a given number of decimal places or significant figures.
- Say which of two measurements is the more accurate.

TEXTBOOK

National	\$18 pp. 111–113, \$19 pp. 114–117
Cambridge	Learner’s Book pp. 74–81
Workbook	Workbook 3.4

01

Explanation

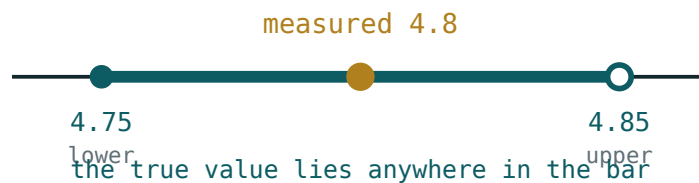
Exact and approximate

A count is exact: there are exactly 27 learners in the class. A measurement is not: a table is “1.2 m long” only to the accuracy of the tape. Every measured number is an **approximate value** of some exact value we shall never know.

DEFINITION

If a is the exact value and x an approximation of it, the **absolute error** is $\Delta = |a - x|$.

We rarely know Δ exactly — we know a **bound** for it. Writing $x = 4.8 \pm 0.05$ says the true value lies between 4.75 and 4.85.



*A measurement of 4.8 rounded to one decimal place:
the true value lies anywhere in the bar.*

Relative error

An error of 1 cm matters enormously on a pencil and not at all on a road. What decides is the error **compared with the size** of the thing measured.

DEFINITION

$$\text{relative error} = \frac{\Delta}{|x|}, \text{ usually written as a percentage}$$

1 cm on a 20 cm pencil is $\frac{1}{20} = 5\%$; 1 cm on a 1000 cm road is 0.1% . The second measurement is a hundred times more accurate, even though the absolute error is the same.

TO COMPARE TWO MEASUREMENTS

Compare the **relative** errors. The smaller relative error is the more accurate measurement.

Rounding

To a decimal place. Look at the first digit you are dropping: 5 or more, round up; less than 5, leave it. $3.147 \rightarrow 3.15$ to 2 d.p.

To significant figures. Count from the first non-zero digit. $0.004736 \rightarrow 0.0047$ to 2 s.f.; $52\,800 \rightarrow 53\,000$ to 2 s.f.

THE ERROR A ROUNDING CREATES

Rounding to the nearest 0.1 gives an absolute error of at most 0.05 — half the last unit. To the nearest whole number, at most 0.5 .

ROUND ONCE, AT THE END

Rounding part-way through a calculation and then again at the end makes the error grow. Keep the full figures until the last line.

02 Worked examples

EXAMPLE 1 The exact value is $a = 3.14159$ and the approximation is $x = 3.14$. Find both errors.

① $\Delta = |3.14159 - 3.14| = 0.00159$
Absolute error.

② $\frac{0.00159}{3.14} \approx 0.00051$
Divide by the approximation.

③ $\approx 0.05\%$
As a percentage.

ANSWER

$\Delta \approx 0.0016$, relative error $\approx 0.05\%$

EXAMPLE

2

A length is measured as 4.8 m to one decimal place. Between which values does the true length lie?

① One decimal place means the nearest 0.1 .

② The error is at most half of that: 0.05 .

③ $4.8 - 0.05 = 4.75$ and $4.8 + 0.05 = 4.85$

④ $4.75 \leq \text{length} < 4.85$

The lower bound is included, the upper is not.

ANSWER

$$4.75 \text{ m} \leq \ell < 4.85 \text{ m}$$

EXAMPLE

3

Which is more accurate: 12 cm measured to the nearest cm, or 250 cm measured to the nearest cm?

① Both have the same absolute error, at most 0.5 cm .

② $\frac{0.5}{12} \approx 4.2\%$

First relative error.

③ $\frac{0.5}{250} = 0.2\%$

Second relative error.

④ The second is far more accurate.

The same error on a bigger quantity matters less.

ANSWER

The 250 cm measurement.

03 Interactive model

Ask for a bound before any calculation — “the answer is somewhere between ...”.

Errors and rounding

Exact $a = 3.14159$, approximation $x = 3.14$.

$$① \quad \Delta = |a - x| = 0.00159$$

Absolute
error

—
just
the
difference,
without
a
sign.

$$② \quad \frac{\Delta}{x} = \frac{0.00159}{3.14} \approx 0.00051$$

Relative
error.

$$③ \quad \approx 0.05\%$$

Multiply
by
100
to
state
it as
a
percentage.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Exact value	Aniq qiymat	Точное значение
Approximate value	Taqribiy qiymat	Приближённое значение
Absolute error	Absolyut xatolik	Абсолютная погрешность
Relative error	Nisbiy xatolik	Относительная погрешность
Round	Yaxlitlash	Округление
Decimal place	O'nli xona	Десятичный знак
Significant figures	Ahamiyatli raqamlar	Значащие цифры
Accuracy	Aniqlik	Точность
Measurement	O'lchash	Измерение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The absolute error of an approximation is:

$$a - x$$

$$|a - x|$$

$$\frac{a}{x}$$

$$a + x$$

A length rounded to 4.8 m (1 d.p.) lies between:

4.7 and 4.9

4.75 and 4.85

4.79 and 4.81

4.8 exactly

0.004736 to 2 s.f. is:

0.00

0.0047

0.0048

0.004

To compare the accuracy of two measurements you use:

- the absolute error
- the relative error
- the larger value
- the number of digits

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Round 3.847 to 1 d.p.

ANSWER 3.8

2. Round 3.847 to 2 d.p.

ANSWER 3.85

3. Round 5629 to 2 s.f.

ANSWER 5600

4. Round 0.00382 to 2 s.f.

ANSWER 0.0038

5. Exact 7, approximation 6.9. Find the absolute error.

ANSWER 0.1

6. Exact 50, approximation 49. Find the absolute error.

ANSWER 1

7. A length is 6 m to the nearest metre. Find the largest possible error.

ANSWER 0.5 m

Medium · the standard question

7 PROBLEMS

1. *Exact 3.14159, approximation 3.14. Find the absolute error.*

ANSWER ≈ 0.0016

2. *The same numbers: find the relative error as a percentage.*

ANSWER $\approx 0.05\%$

3. *A length reads 4.8 m to 1 d.p. Give the bounds.*

ANSWER $4.75 \leq \ell < 4.85$

4. *A mass reads 250 g to the nearest 10 g. Give the bounds.*

ANSWER $245 \leq m < 255$

5. *Exact 20, approximation 19.5. Find the relative error.*

ANSWER 2.5%

6. *Which is more accurate: 12 cm or 250 cm, both to the nearest cm?*

ANSWER 250 cm — relative error 0.2% against 4.2%.

7. *Round 0.0004736 to 3 s.f.*

ANSWER 0.000474

Hard · stretch and reason

7 PROBLEMS

1. A rectangle is 4.2 m by 3.5 m, both to 1 d.p. Find the bounds of its perimeter.

ANSWER $4.15 \leq a < 4.25$, $3.45 \leq b < 3.55$, so $15.2 \leq P < 15.6$

2. The same rectangle: find the bounds of its area.

ANSWER $14.3175 \leq S < 15.0875$

3. A measurement has absolute error at most 0.02 and relative error 0.5%. Estimate the value.

ANSWER $\frac{0.02}{x} = 0.005$, so $x = 4$

4. Two masses: $3.5 \text{ kg} \pm 0.05$ and $480 \text{ g} \pm 5 \text{ g}$. Which is more accurate?

ANSWER 1.43% against 1.04% — the 480 g measurement.

5. Explain why rounding twice can be worse than rounding once.

ANSWER Each rounding adds its own error, and the errors accumulate: $2.46 \rightarrow 2.5 \rightarrow 3$ but $2.46 \rightarrow 2$ directly.

6. A number rounds to 7.30 to 2 d.p. Give its bounds.

ANSWER $7.295 \leq x < 7.305$

7. A speed of 60 km/h is measured to the nearest 5 km/h. Find the relative error at most.

ANSWER $\frac{2.5}{60} \approx 4.2\%$

07 Homework

Homework — 6 problems

Algebra 8, §§18–19, pp. 111–117.

- Round 2.7182 to 1, 2 and 3 decimal places.
- Round 34 782 to 2 and to 3 significant figures.
- Exact 8, approximation 7.6. Find the absolute and relative errors.
- A length reads 12.3 cm to 1 d.p. Give the bounds.

5. *Which is more accurate: 5 m to the nearest metre, or 500 m to the nearest metre?*
6. *A square has side 6.4 cm to 1 d.p. Find the bounds of its perimeter.*